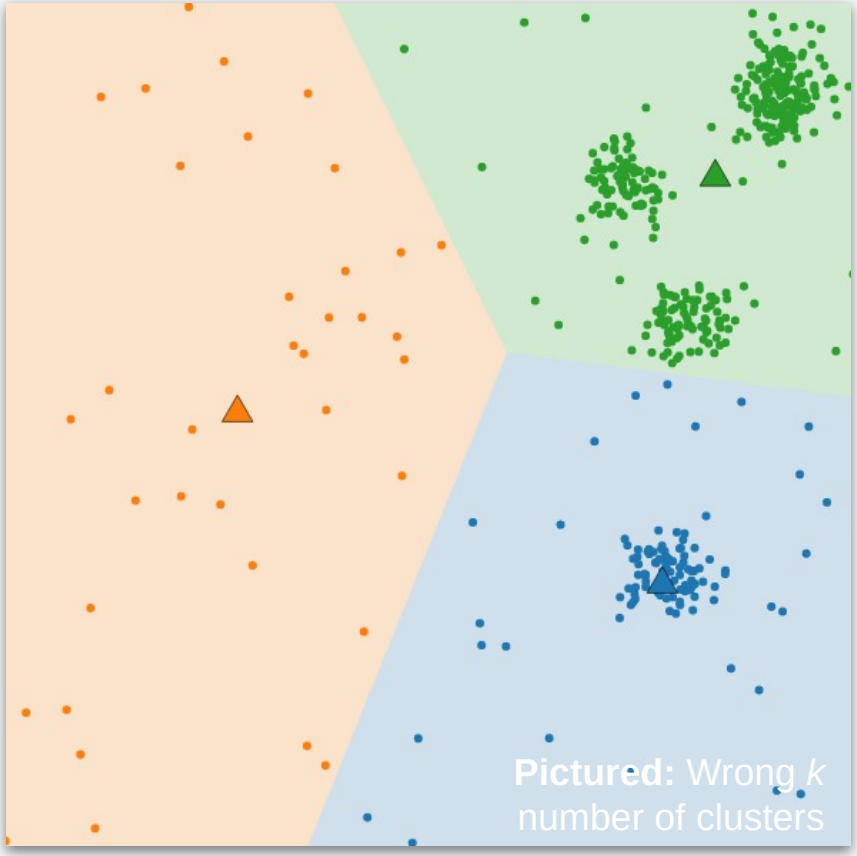
Autoencoders

Recap of **K-Means**

Goal:

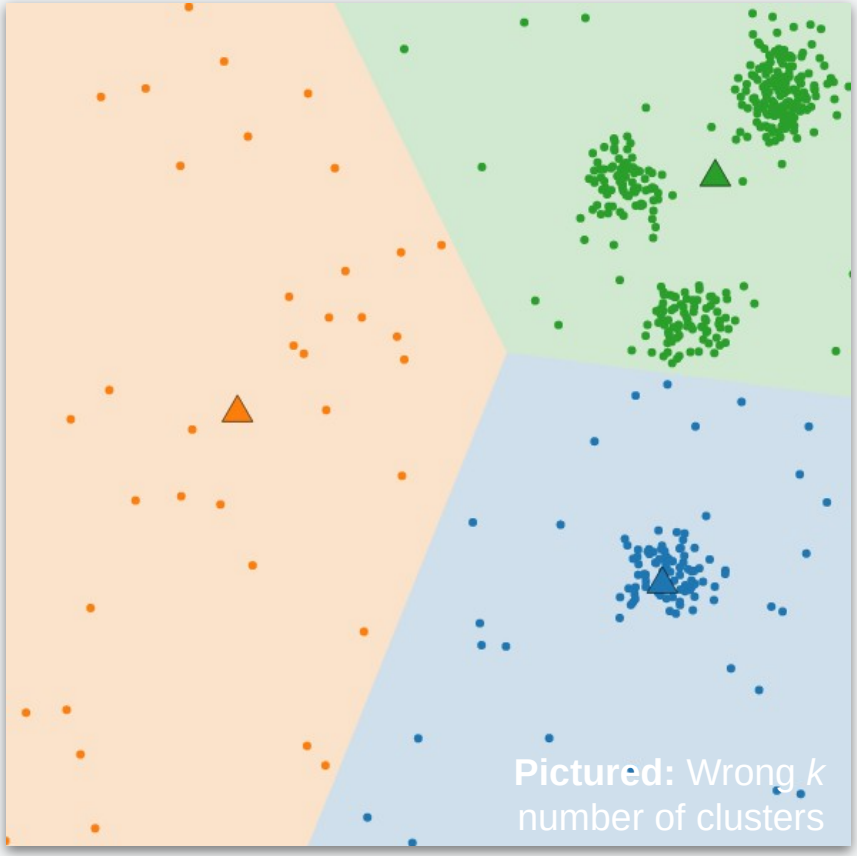
Partition *n* observations into *k* clusters



When **K-Means** Fails?

- Scales poorly with **large numbers** of observations

- **“Curse of Dimensionality”**

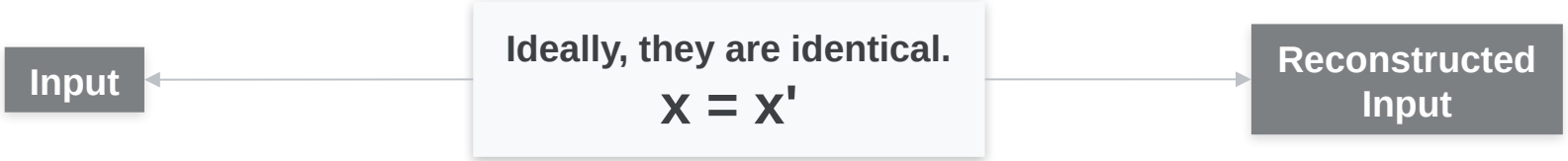



Input
Prediction
Error

You don’t know what you **don’t know**.

You don’t know what you **don’t know**.

But **you do know** what **you know**.


Input
Ideally, they are identical.
x = x'
Reconstructed Input

Input

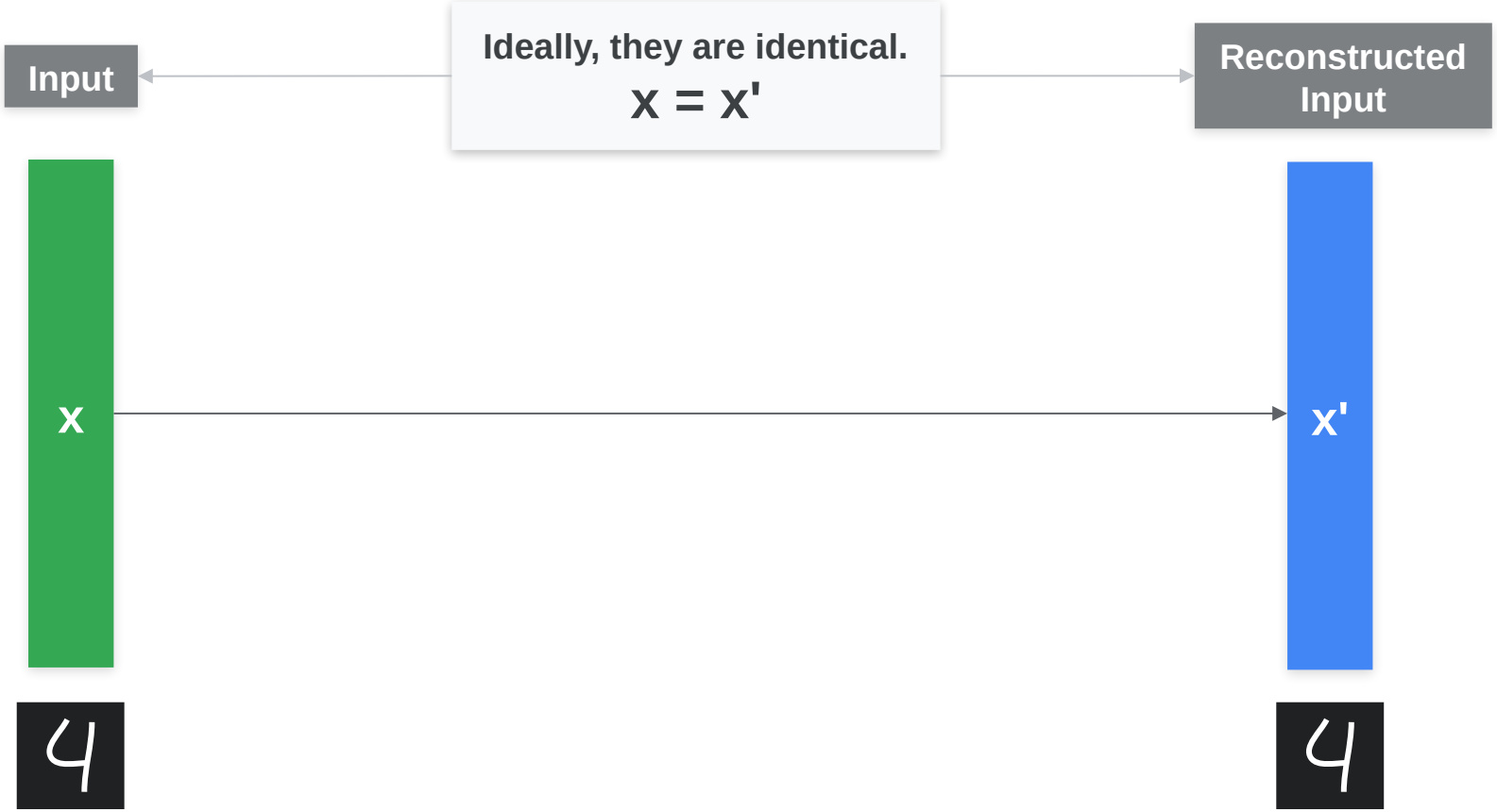
Ideally, they are identical.

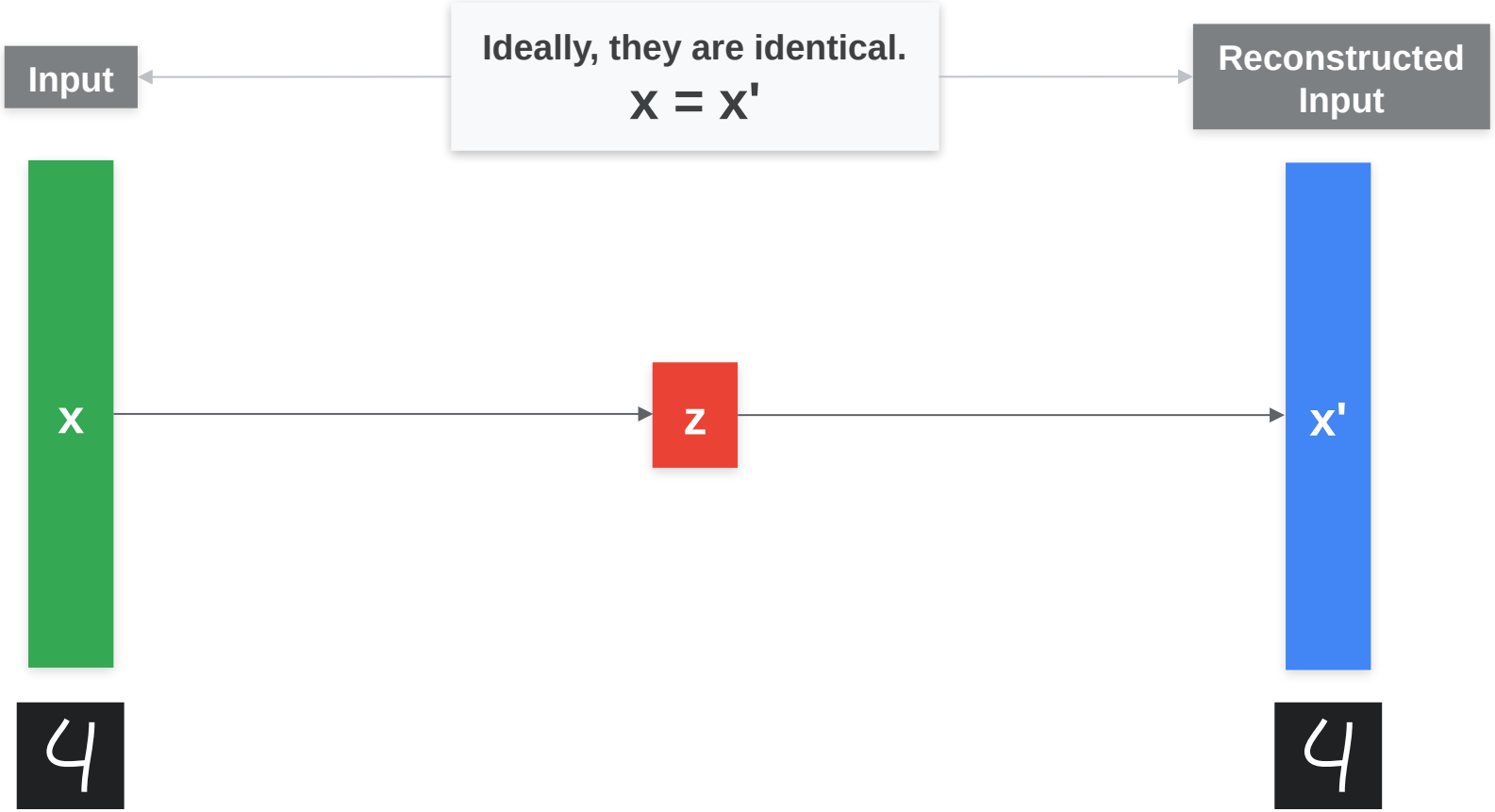
$x = x'$

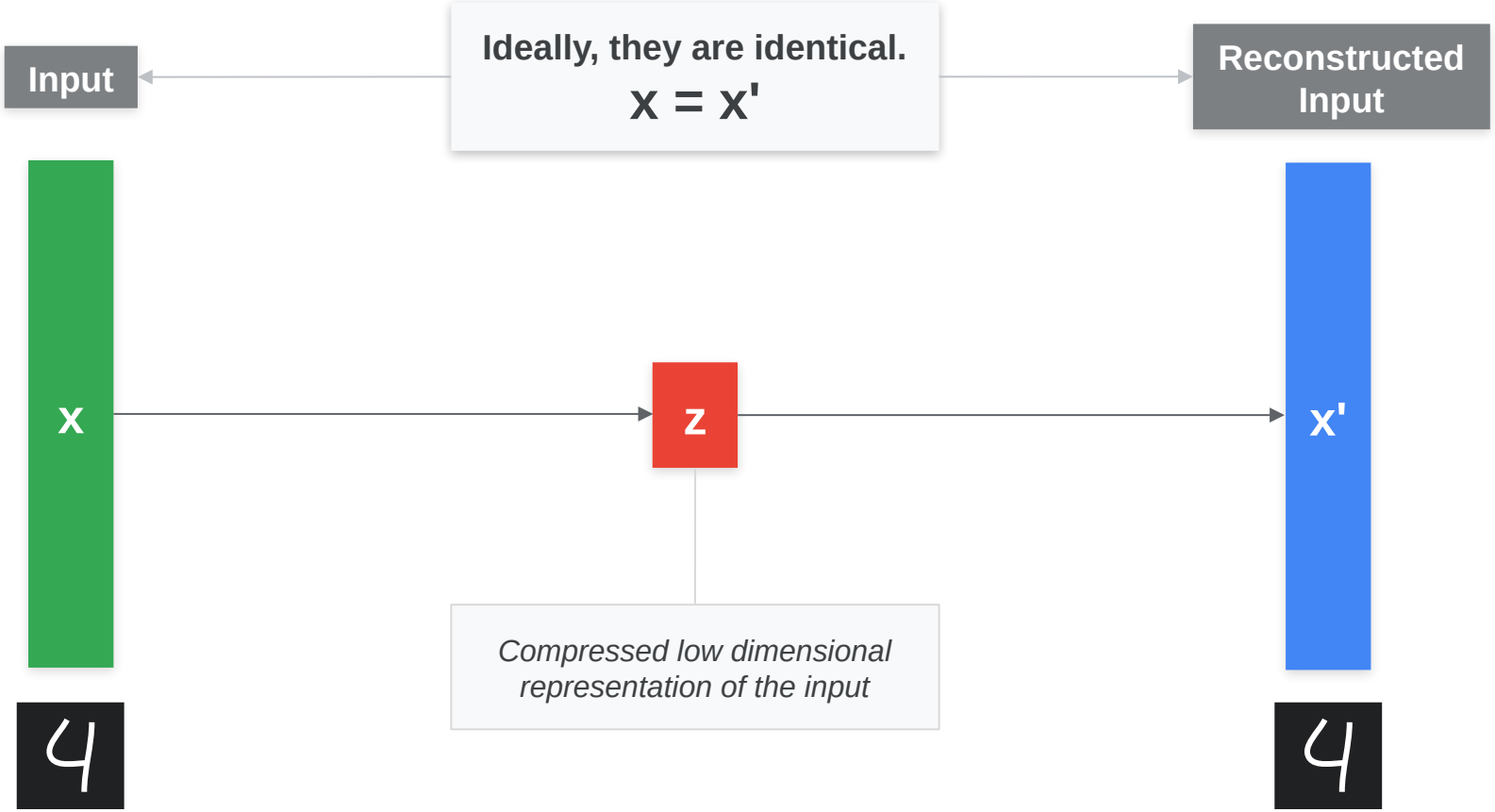
Reconstructed Input

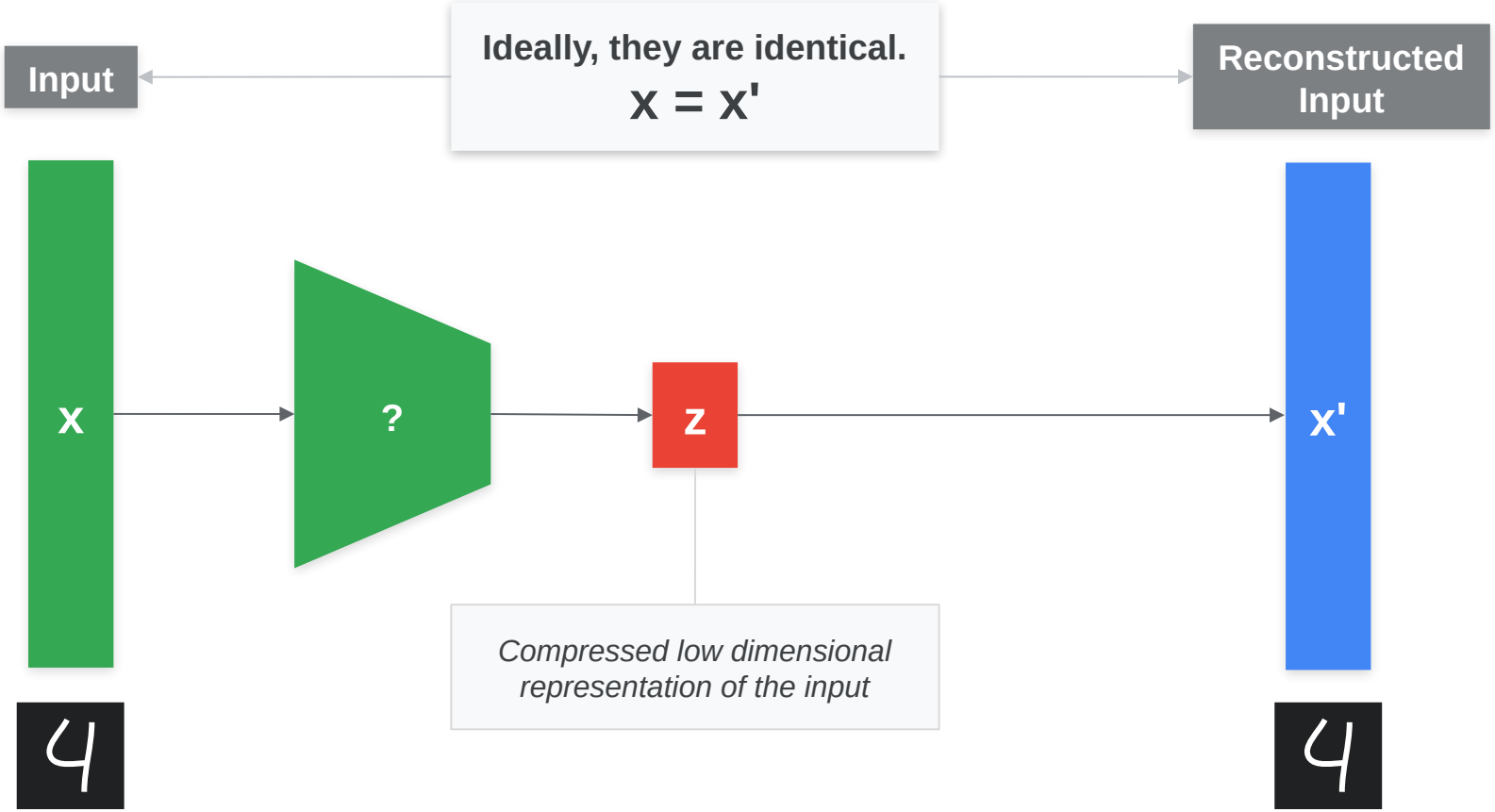


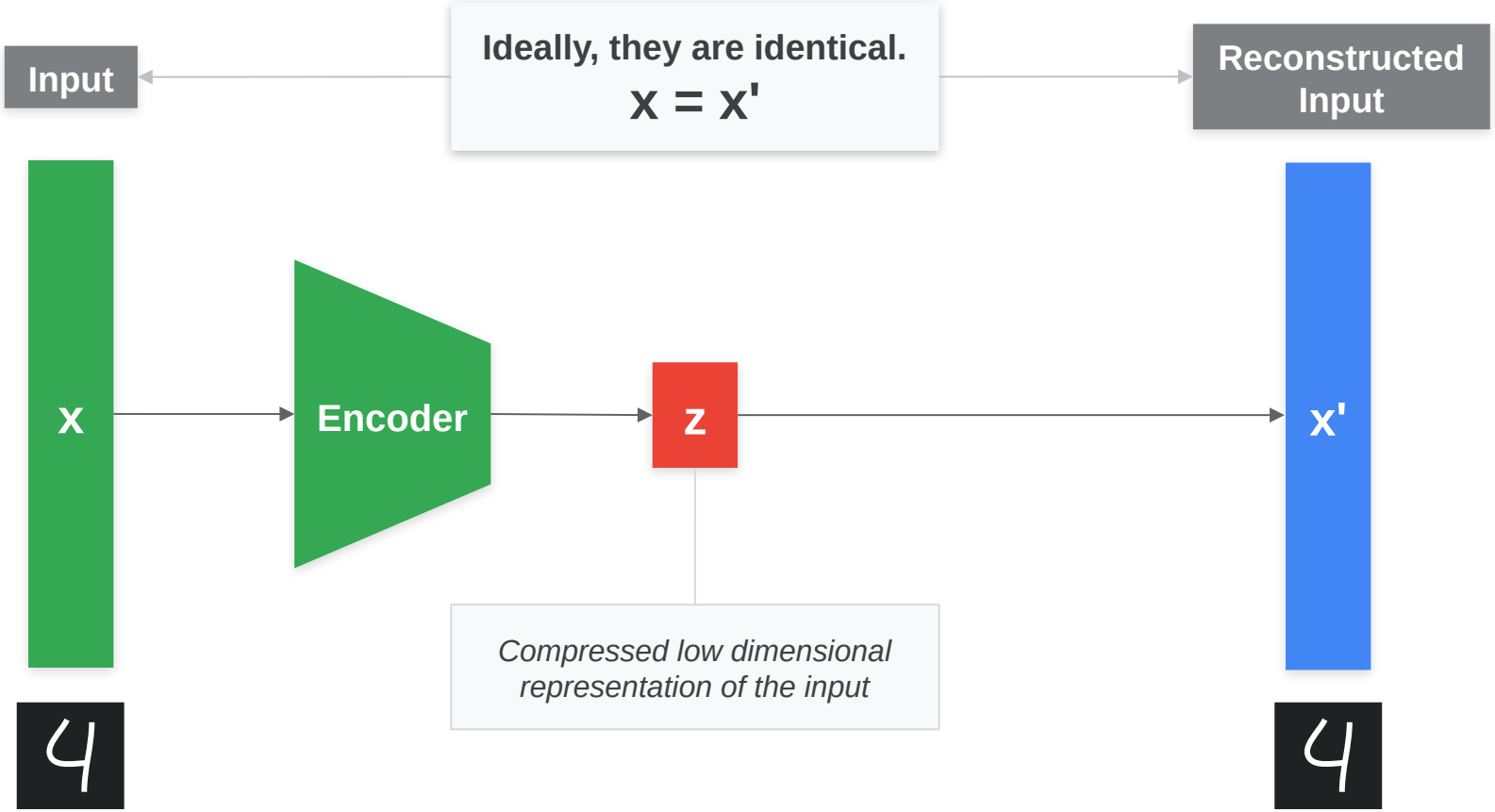


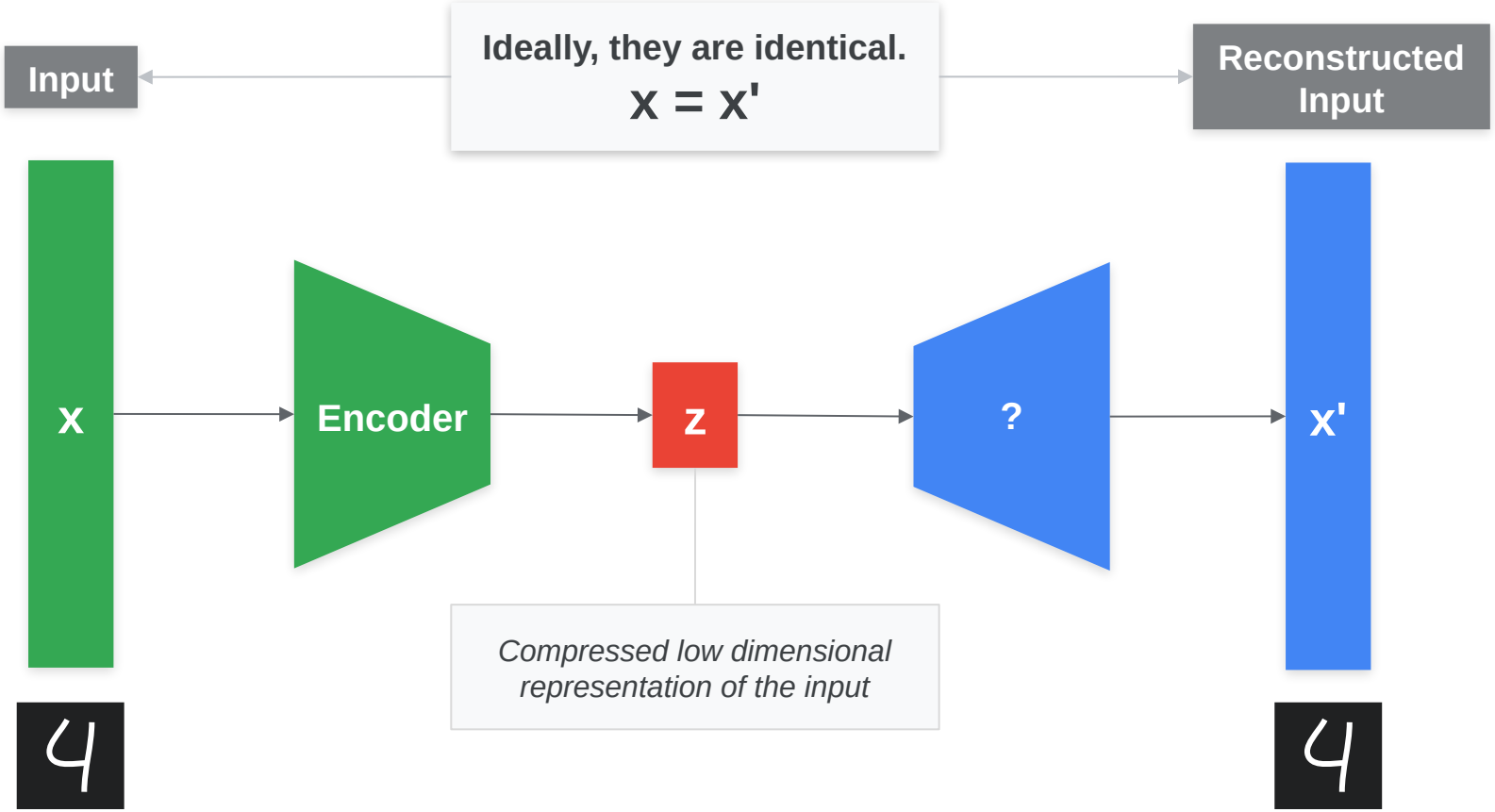

Input
Ideally, they are identical.
x = x'
Reconstructed Input
x
x'

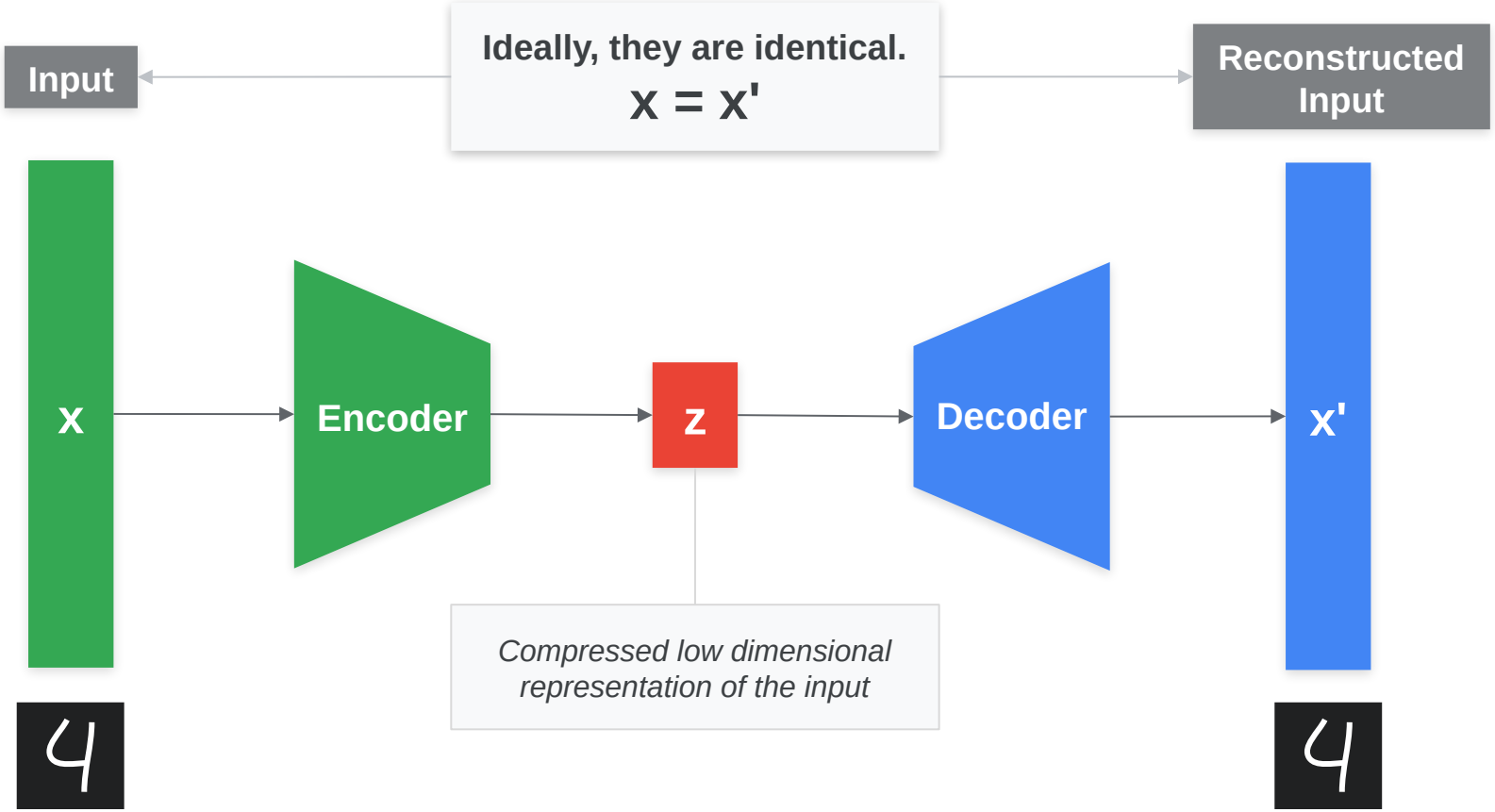

Input
Ideally, they are identical.
x = x'
Reconstructed Input
x
z
x'

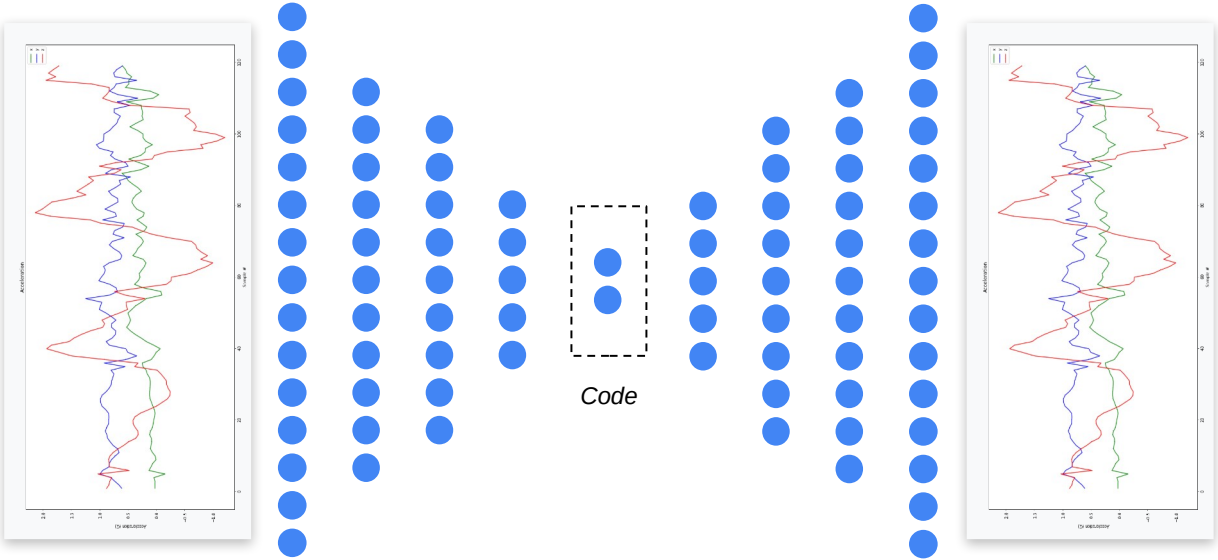

Input
Ideally, they are identical.
x = x'
Reconstructed Input
x
z
x'
Compressed low dimensional representation of the input

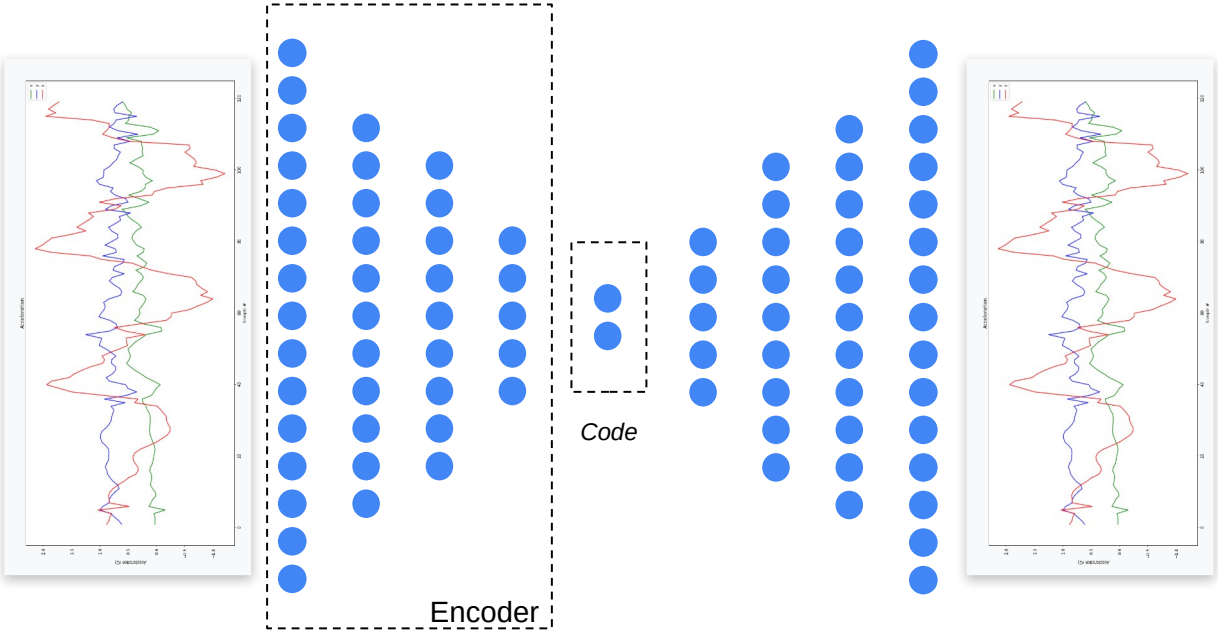

Input
Ideally, they are identical.
x = x'
Reconstructed Input
x
?
z
x'
Compressed low dimensional representation of the input

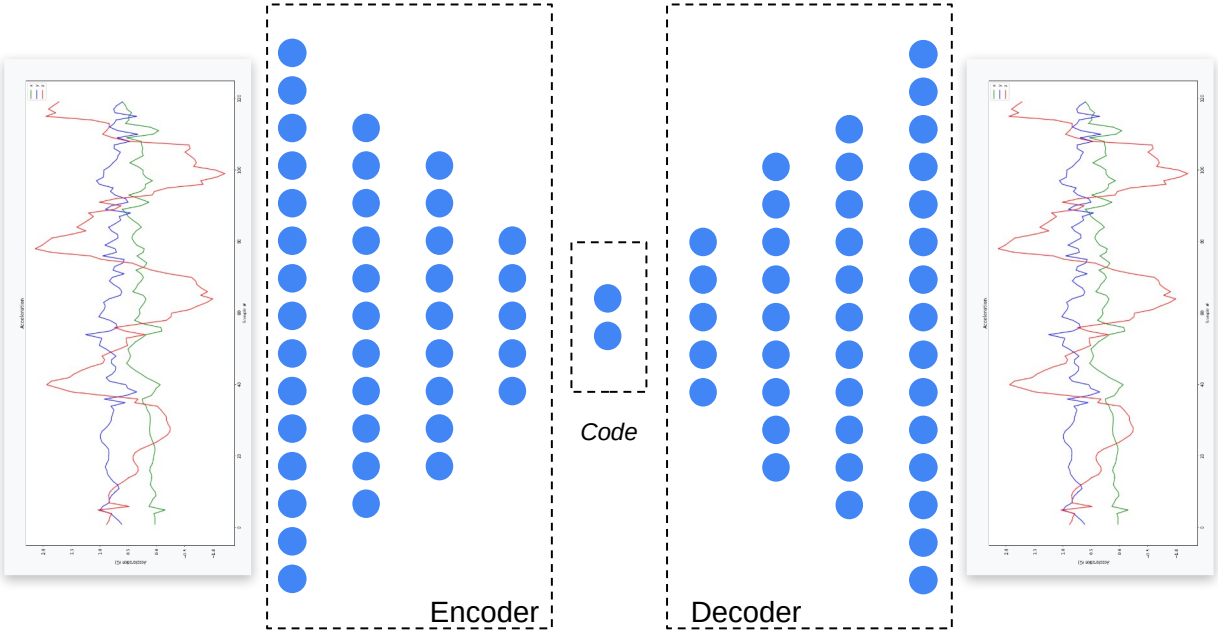

Input
Ideally, they are identical.
x = x'
Reconstructed Input
x
Encoder
z
x'
Compressed low dimensional representation of the input

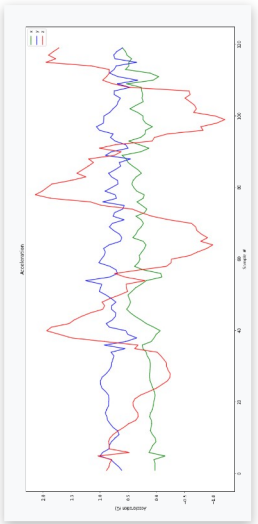

Input
Ideally, they are identical.
x = x'
Reconstructed Input
x
Encoder
z
?
x'
Compressed low dimensional representation of the input

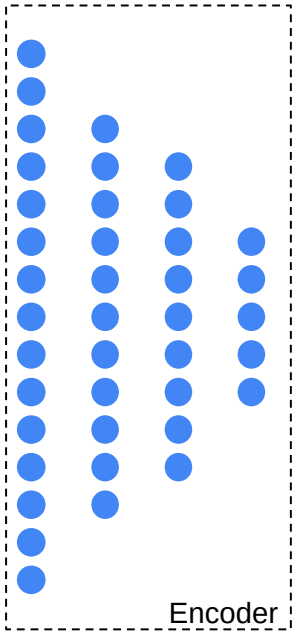

Input
Ideally, they are identical.
x = x'
Reconstructed Input
x
Encoder
z
Decoder
x'
Compressed low dimensional representation of the input


Code

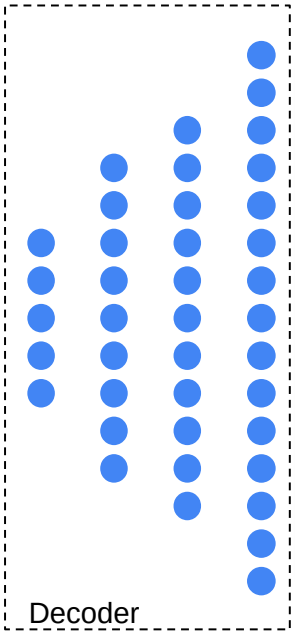

Encoder
Code

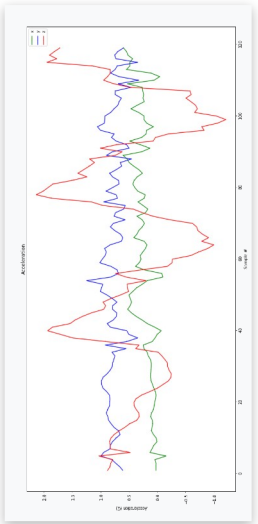

Encoder
Code
Decoder



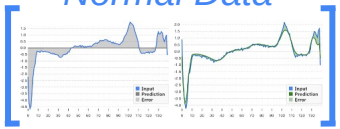


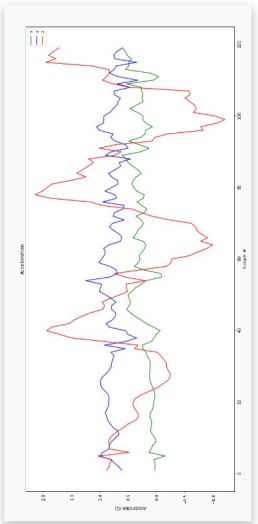


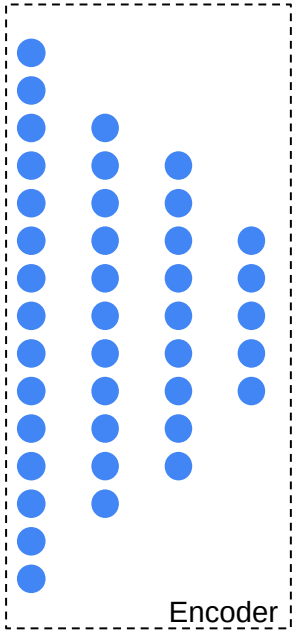




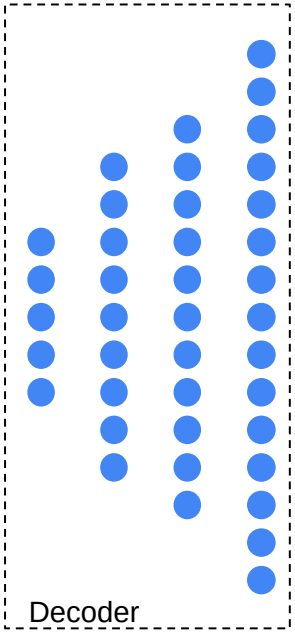
Normal Data

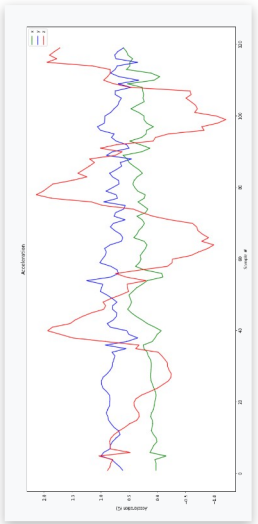




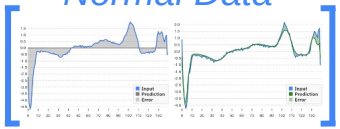




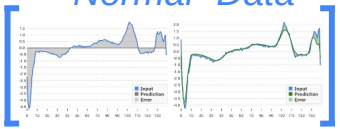


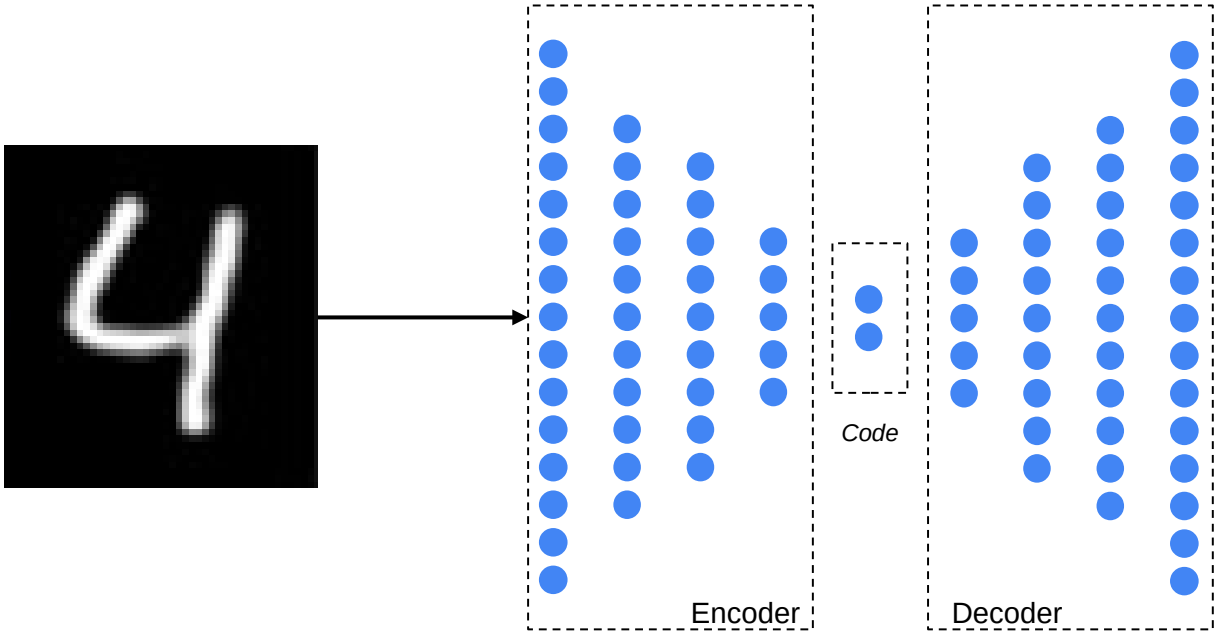


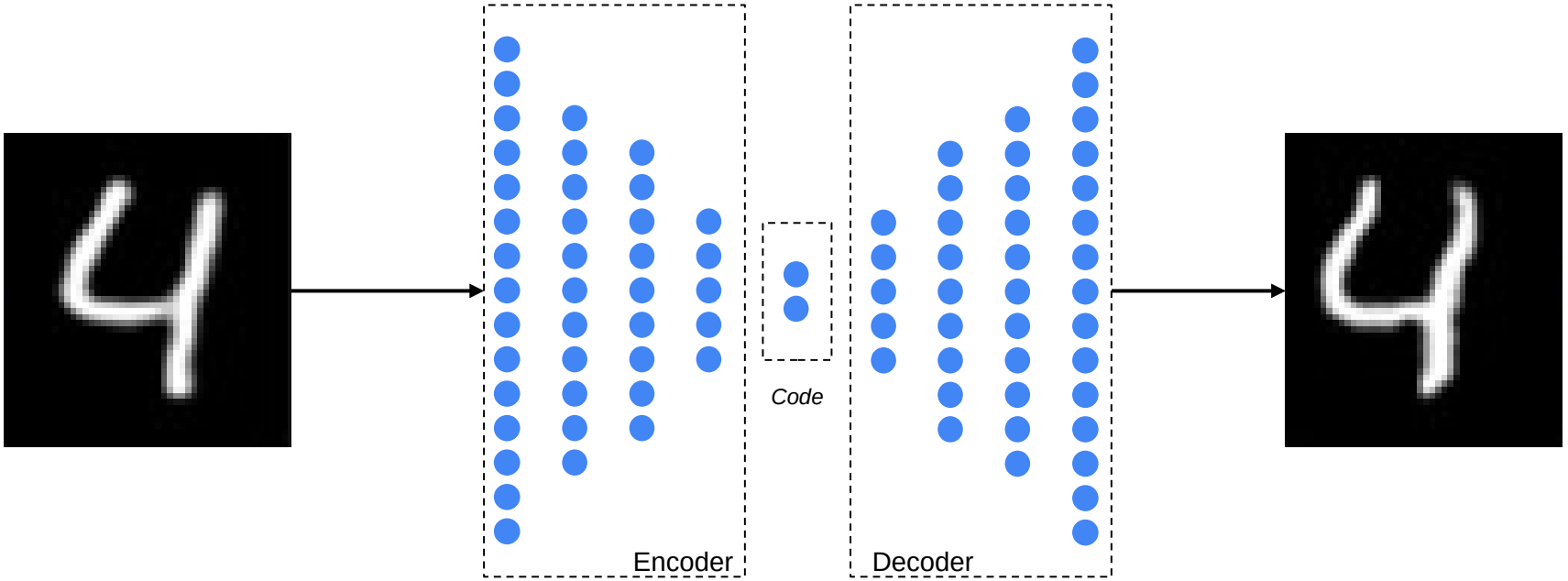
Normal Data

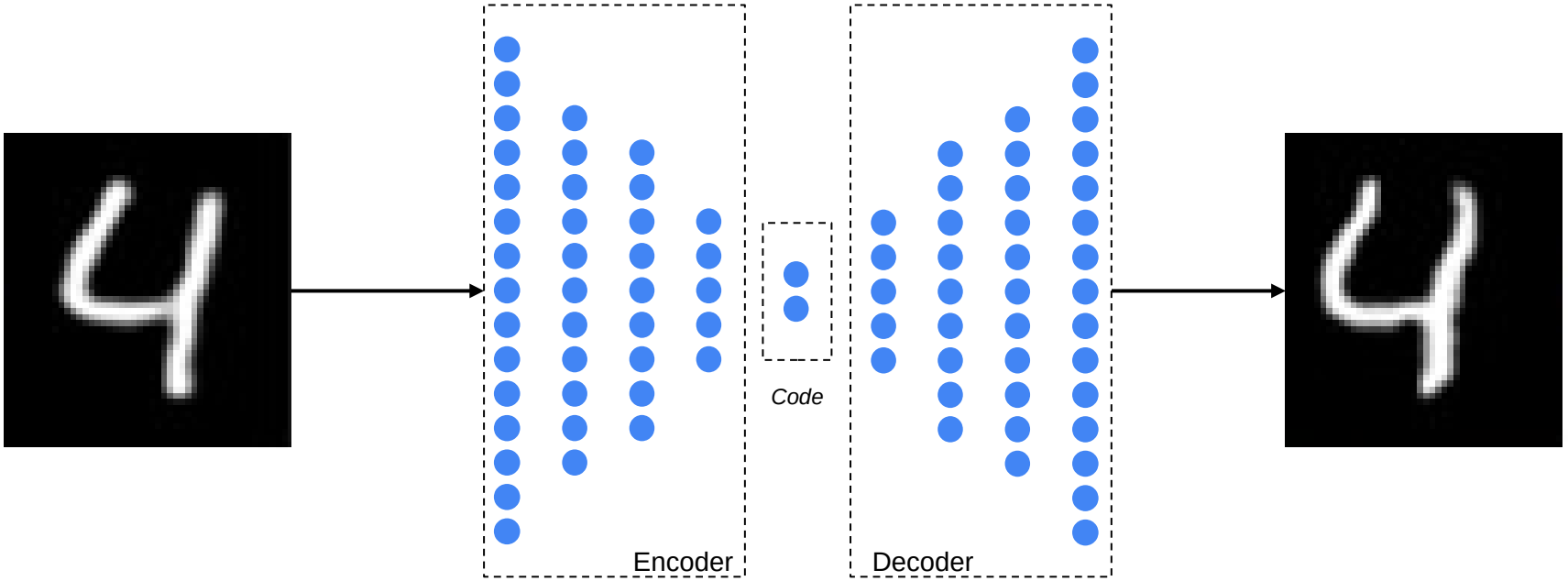


“Normal” Data



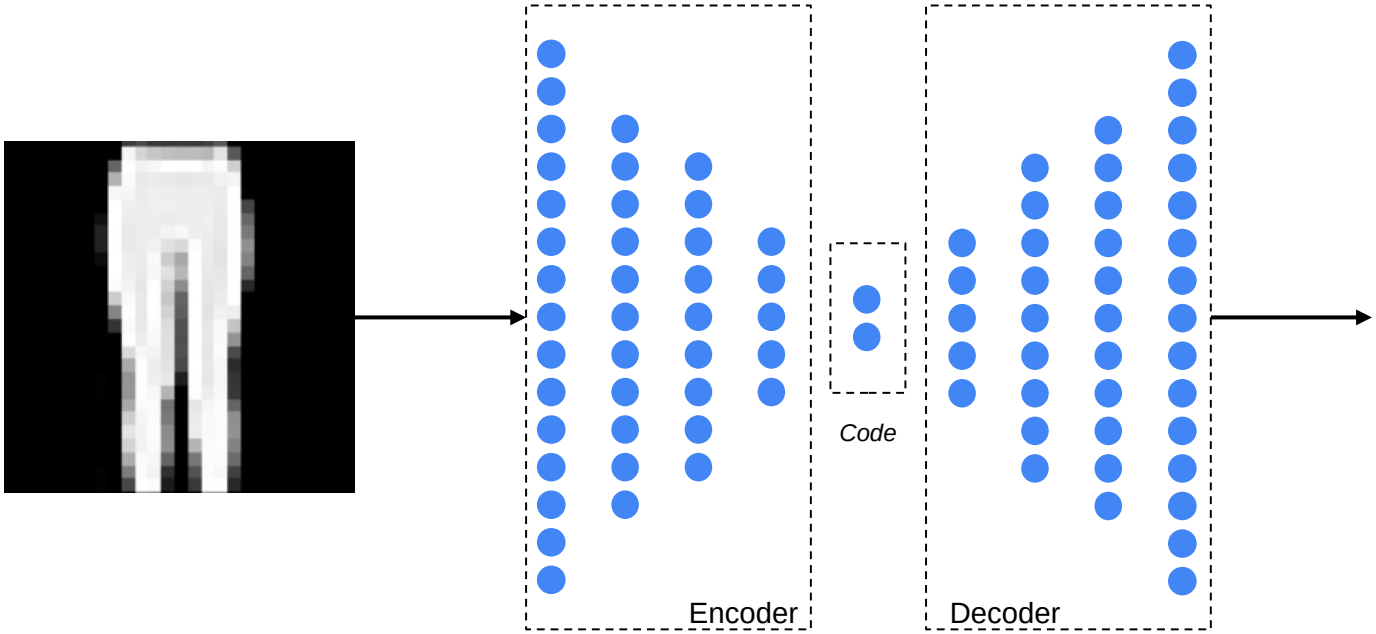

Encoder
Code
Decoder

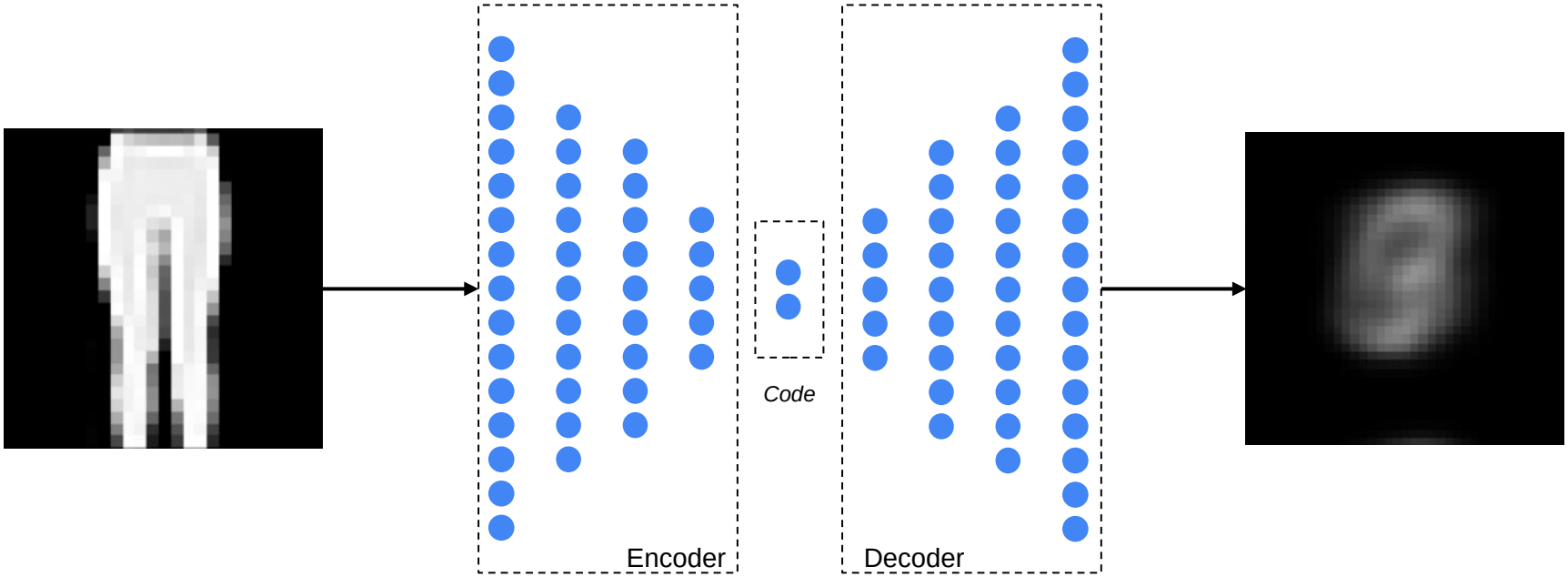

Encoder
Code
Decoder



Low Reconstruction Loss

(Low Mean Square Error)


Encoder
Code
Decoder

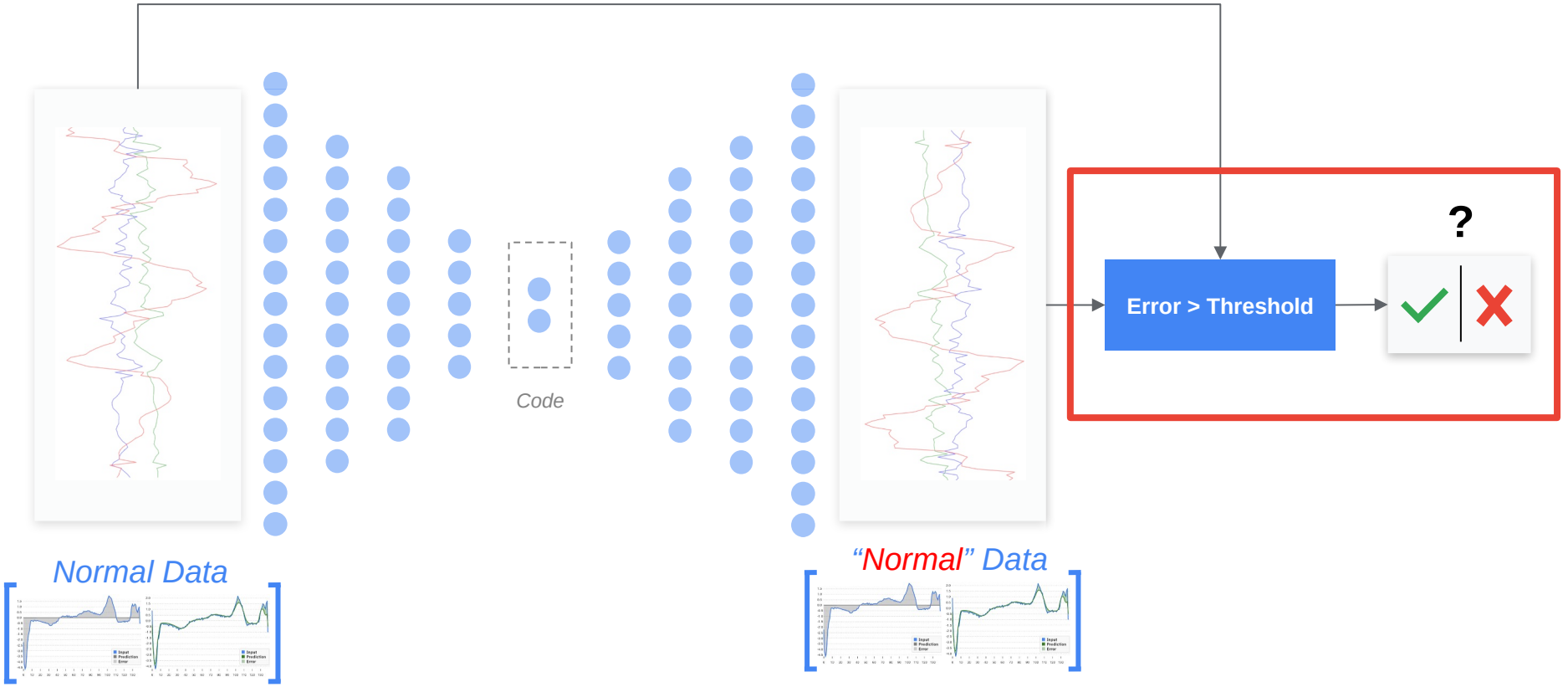


Low Reconstruction Loss

(Low Mean Square Error)

Unbalanced Data




Code
Error > Threshold
?
Normal Data
"Normal" Data

Fullscreen
Show Presenter